PYTHON FOR HEALTHCARE · STANFORD TECHNOLOGY TRAINING

Python You Need to Read

Session 1 handout · Read it, run it, check it

An assistant can write this code for you; you still have to read it. Here's what each common pattern does, and the mistake it tends to hide.

Pattern	What it does	Watch for
LOOKING		
`df.shape`	(rows, columns)	Rows that appeared or vanished
`df.describe()`	Count, mean, min, max of number columns	A maximum of 999 is a code, not a reading
`df["arm"].value_counts()`	How often each value appears	Near-duplicates such as "Coaching " and "coaching"
`df.isna().sum()`	Blanks in each column	Blanks can be real missing data, not errors
SELECTING		
`df[df["sbp"] >= 140]`	Keep rows that meet a condition	== compares; a single = stores
`df.loc[rows, columns]`	Rows by a condition, columns by name	Check how many rows are left
CLEANING		
`df.drop_duplicates()`	Remove exact duplicate rows	The result must be saved back: `df = ...`
`.str.strip().str.lower()`	Tidy text values	Changes every value in the column
`.replace(999, np.nan)`	Turn a code into a true blank	Check every column that uses the code
`df.dropna()`	Drop rows with **any** blank	Drops people for a blank in any column
`df.dropna(subset=["sbp_w12"])`	Drop rows missing one column	Usually the right choice for an outcome
COMBINING		
`df.groupby("arm")["sbp"].mean()`	One summary per group	Look at `count` too
`a.merge(b, on="id", validate="one_to_one")`	Join two tables	Without `validate`, a merge can multiply rows
`df.pivot(index=, columns=, values=)`	Long to wide: one row per person	Check the new shape
CHECKING AND MODELING		
`assert condition, "message"`	Stop if the condition is false	Your stop rules
`smf.ols("y ~ arm + x", data=df).fit()`	Linear regression	Rows with a blank in any model column are dropped silently
`C(arm, Treatment(reference="usual care"))`	Set the reference group	The default reference is alphabetical
`model.conf_int()`	95% confidence intervals	Report with every estimate
`stats.ttest_ind(a, b, equal_var=False)`	Welch's t-test	Leaving out `equal_var=False` gives Student's t-test

Five questions for any cell, yours or the assistant's

1. What goes in? Which table, which columns? **2. Which rows are kept or dropped?** Filters, `dropna()`, `drop_duplicates()`. **3. Does the row count change, and should it?** `merge`, `groupby`, `pivot`. **4. What's computed, and on which rows? 5. Does the output match your prediction?** If not, find out why.

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Check It Before You Trust It

Six checks for any analysis, whoever wrote the code

Check	PASS if	How to check
1. Counts	Table 1 shows everyone randomized, and every model reports its n	Count rows by arm yourself, and count the rows that have the outcome
2. Nobody dropped for the wrong reason	Models use everyone who has the outcome	Search the code for `dropna()` without `subset=`; compare the model's n with your count
3. Direction	The effect is labeled (A minus B), and its sign matches the raw means	Compare the group means; watch which group is the reference
4. The model matches the plan	Right outcome, covariates, reference group, and test	Read the formula and the test options line by line
5. Effect, interval, units	The estimate comes with a 95% CI, in real units	A p-value alone isn't a result
6. The figure agrees	The chart's values match the table	Recompute two or three plotted numbers

Before you share any result

- ☐ **Restart and run all** (Runtime → Restart session and run all). If it doesn't run top to bottom, it isn't a result yet.
- ☐ **Write the plan before you ask.** A vague request lets the assistant choose your outcome, population, and model. A written plan makes those choices yours. That's what *context is the lever* means.
- ☐ **Keep your stop rules.** They make the notebook refuse to continue when the data break an assumption.
- ☐ **Write down every cleaning decision**, with how many rows it touched and why.

Where real data can go at Stanford

From UIT's AI matrix and its Nero service page, checked October 21, 2026. Rules change: check **uit.stanford.edu/ai/matrix** before you use real data, and follow your school's or department's own rules. Only UIT-managed workspaces count, signed in with Stanford SSO.

Tool	Approved for	Notes
Google Colab and Gemini on a personal account	Not a Stanford service	What we use in class, with synthetic data only
Stanford AI Playground	High Risk, no PHI	Approved May 13, 2026; PHI review in progress
Claude for Education, Claude Code	High Risk, no PHI	UIT-managed workspace
ChatGPT Edu, ChatGPT Codex	High Risk, including PHI	UIT-managed workspace
Google Gemini Chat; Gemini Enterprise	High Risk, including PHI	Signed in with your Stanford account (not Gemini in Colab on a personal account)
SHC–SoM Secure GPT	High Risk, including PHI	School of Medicine only
Nero; Carina 2.0	High Risk, including PHI	Research computing; Nero needs a PI sponsor and a Data Risk Assessment